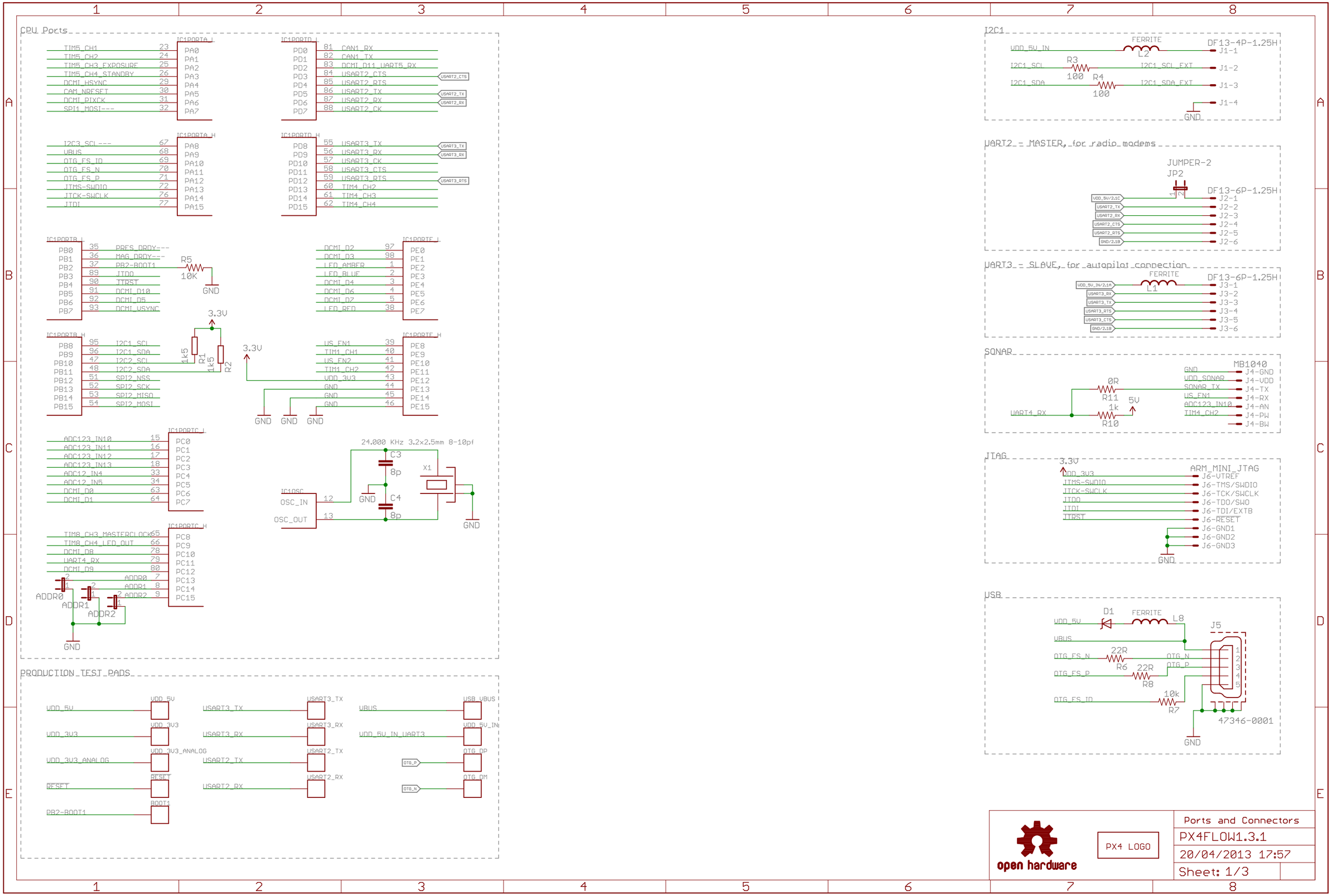
CPU Ports
IC1PORTA_L
TIM5_CH1 23 PA0
TIM5_CH2 24 PA1
TIM5_CH3_EXPOSURE 25 PA2
TIM5_CH4_STANDBY 26 PA3
DCMI_HSYNC 29 PA4
CAM_NRESET 30 PA5
DCMI_PIXCK 31 PA6
SPI1_MOSI--- 32 PA7
IC1PORTD_L
PD0 81 CAN1_RX
PD1 82 CAN1_TX
PD2 83 DCMI_D11_UART5_RX
PD3 84 USART2_CTS
PD4 85 USART2_RTS
PD5 86 USART2_TX
PD6 87 USART2_RX
PD7 88 USART2_CK
IC1PORTA_H
I2C3_SCL--- 67 PA8
VBUS 68 PA9
OTG_FS_ID 69 PA10
OTG_FS_N 70 PA11
OTG_FS_P 71 PA12
JTMS-SWDIO 72 PA13
JTCK-SWCLK 76 PA14
JTDI 77 PA15
IC1PORTD_H
PD8 55 USART3_TX
PD9 56 USART3_RX
PD10 57 USART3_CK
PD11 58 USART3_CTS
PD12 59 USART3_RTS
PD13 60 TIM4_CH2
PD14 61 TIM4_CH3
PD15 62 TIM4_CH4
IC1PORTB_L
PB0 35 PRES_DRDY---
PB1 36 MAG_DRDY---
PB2 37 PB2-BOOT1
PB3 89 JTDO
PB4 90 JTDST
PB5 91 DCMI_D10
PB6 92 DCMI_D5
PB7 93 DCMI_VSYNC
R5 10K
GND
IC1PORTE_L
DCMI_D2 97 PE0
DCMI_D3 98 PE1
LED_AMBER 1 PE2
LED_BLUE 2 PE3
DCMI_D4 3 PE4
DCMI_D6 4 PE5
DCMI_D7 5 PE6
LED_RED 38 PE7
3.3V
IC1PORTB_H
PB8 95 I2C1_SCL
PB9 96 I2C1_SDA
PB10 47 I2C2_SCL
PB11 48 I2C2_SDA
PB12 51 SPI2_NSS
PB13 52 SPI2_SCK
PB14 53 SPI2_MISO
PB15 54 SPI2_MOSI
1k5 R1
1k5 R2
3.3V
IC1PORTE_H
US_EN1 39 PE8
TIM1_CH1 40 PE9
US_EN2 41 PE10
TIM1_CH2 42 PE11
VDD_3V3 43 PE12
GND 44 PE13
GND 45 PE14
GND 46 PE15
GND GND GND
IC1PORTC_L
ADC123_IN10 15 PC0
ADC123_IN11 16 PC1
ADC123_IN12 17 PC2
ADC123_IN13 18 PC3
ADC12_IN4 33 PC4
ADC12_IN5 34 PC5
DCMI_D0 63 PC6
DCMI_D1 64 PC7
24.000 KHz 3.2x2.5mm 8-10pf
C3 8p
X1
IC1OSC
OSC_IN 12
OSC_OUT 13
GND C4 8p
GND
IC1PORTC_H
TIM8_CH3_MASTERCLOCK 65 PC8
TIM8_CH4_LED_OUT 66 PC9
DCMI_D8 78 PC10
UART4_RX 79 PC11
DCMI_D9 80 PC12
ADDR0 7 PC13
ADDR1 8 PC14
ADDR2 9 PC15
ADDR0
ADDR1
ADDR2
GND
PRODUCTION_TEST_PADS
VDD_5V VDD_5V
VDD_3V3 VDD_3V3
VDD_3V3_ANALOG VDD_3V3_ANALOG
RESET RESET
PB2-BOOT1 BOOT1
USART3_TX USART3_TX
USART3_RX USART3_RX
USART2_TX USART2_TX
USART2_RX USART2_RX
VBUS USB_VBUS
VDD_5V_IN_UART3 VDD_5V_IN
OTG_P OTG_DP
OTG_N OTG_DM
I2C1
VDD_5V_IN FERRITE L2 DF13-4P-1.25H J1-1
I2C1_SCL R3 100 I2C1_SCL_EXT J1-2
I2C1_SDA R4 100 I2C1_SDA_EXT J1-3
J1-4
GND
UART2 - MASTER, for radio modems
JUMPER-2
JP2
DF13-6P-1.25H
VDD_5V/2.1C J2-1
USART2_TX J2-2
USART2_RX J2-3
USART2_CTS J2-4
USART2_RTS J2-5
GND/2.1B J2-6
UART3 - SLAVE, for autopilot connection
FERRITE L1
DF13-6P-1.25H
VDD_5V_IN/2.1A J3-1
USART3_RX J3-2
USART3_TX J3-3
USART3_RTS J3-4
USART3_CTS J3-5
GND/2.1B J3-6
SONAR
MB1040
GND J4-GND
VDD_SONAR J4-VDD
SONAR_TX J4-TX
US_EN1 J4-RX
ADC123_IN10 J4-AN
TIM4_CH2 J4-PW
J4-BW
0R R11
5V
UART4_RX
1k R10
JTAG
3.3V
ARM_MINI_JTAG
VDD_3V3 J6-VTREF
JTMS-SWDIO J6-TMS/SWDIO
JTCK-SWCLK J6-TCK/SWCLK
JTDO J6-TDO/SWO
JTDI J6-TDI/EXTB
JTRST J6-RESET
J6-GND1
J6-GND2
J6-GND3
GND
USB
VDD_5V D1 FERRITE L8
J5
VBUS
OTG_FS_N 22R R6 OTG_N
OTG_FS_P 22R R8 OTG_P
OTG_FS_ID 10k R7
47346-0001
GND
open hardware
PX4 LOGO
Ports and Connectors
PX4FLOW1.3.1
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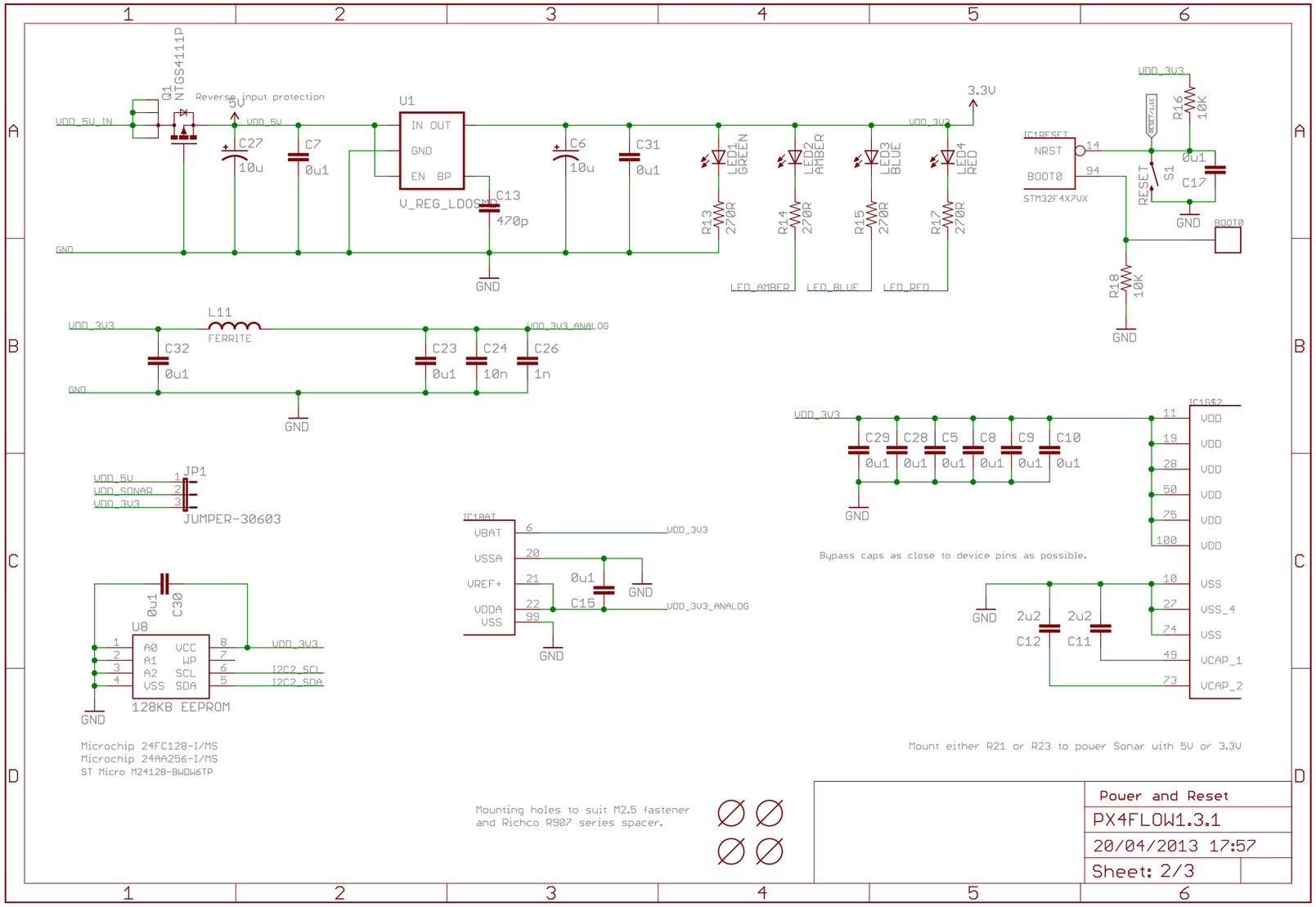

Power and Reset	
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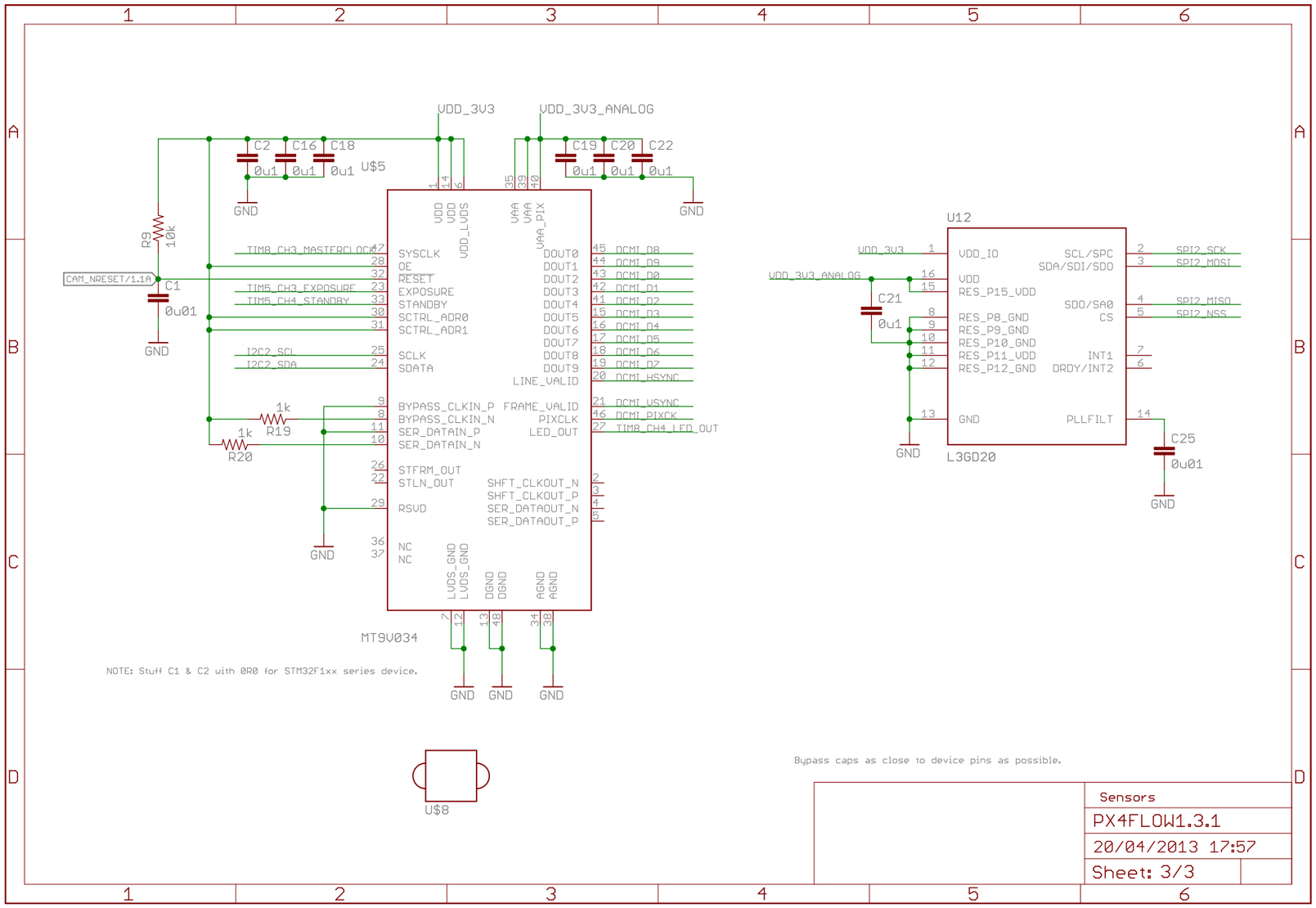
VDD_3V3
VDD_3V3_ANALOG
C2 C16 C18
0u1 0u1 0u1
U$5
C19 C20 C22
0u1 0u1 0u1
GND
R9
10k
CAM_NRESET/1.1A
C1
0u01
TIM8_CH3_MASTERCLOCK
TIM5_CH3_EXPOSURE
TIM5_CH4_STANDBY
I2C2_SCL
I2C2_SDA
1k
R19
1k
R20
SYSCLK
OE
RESET
EXPOSURE
STANDBY
SCTRL_ADR0
SCTRL_ADR1
SCLK
SDATA
BYPASS_CLKIN_P
BYPASS_CLKIN_N
SER_DATAIN_P
SER_DATAIN_N
STFRM_OUT
STLN_OUT
RSVD
NC
NC
VDD
VDD
VDD_LVDS
VAA
VAA
VAA_PIX
DOUT0
DOUT1
DOUT2
DOUT3
DOUT4
DOUT5
DOUT6
DOUT7
DOUT8
DOUT9
LINE_VALID
FRAME_VALID
PIXCLK
LED_OUT
SHFT_CLKOUT_N
SHFT_CLKOUT_P
SER_DATAOUT_N
SER_DATAOUT_P
LVDS_GND
LVDS_GND
DGND
DGND
AGND
AGND
DCMI_D8
DCMI_D9
DCMI_D0
DCMI_D1
DCMI_D2
DCMI_D3
DCMI_D4
DCMI_D5
DCMI_D6
DCMI_D7
DCMI_HSYNC
DCMI_VSYNC
DCMI_PIXCK
TIM8_CH4_LED_OUT
MT9V034
NOTE: Stuff C1 & C2 with 0R0 for STM32F1xx series device.
U12
VDD_3V3
VDD_3V3_ANALOG
C21
0u1
VDD_IO
VDD
RES_P15_VDD
RES_P8_GND
RES_P9_GND
RES_P10_GND
RES_P11_VDD
RES_P12_GND
GND
SCL/SPC
SDA/SDI/SDO
SDO/SA0
CS
INT1
DRDY/INT2
PLLFILT
SPI2_SCK
SPI2_MOSI
SPI2_MISO
SPI2_NSS
C25
0u01
L3GD20
U$8
Bypass caps as close to device pins as possible.
Sensors
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